

WASP Suite: Workload Analysis for Scalable Processors

Final Results from a Full Core-Count Sweep for Scheduling and Microarchitecture Analysis

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Abstract—This paper presents the Workload Analysis for Scalable Processors (WASP) Suite, a comprehensive C-based framework for analysing multicore processor performance, scheduling efficiency, and microarchitecture behaviour using real measured latency rather than synthetic latency equations. WASP integrates load generation with `Linux perf_event_open()` counters, records transaction-level timing via `clock_gettime(CLOCK_MONOTONIC)`, and produces percentile-oriented performance analysis (P50/P95/P99). Final evaluation completes the full benchmark grid: four workloads (CPU, memory, I/O, mixed), three intensities (25%, 50%, 75%), four active core counts (1, 2, 4, 8), and five repetitions per cell (240 total runs). With normalized operation definitions, workload means are 0.0583 μ s (CPU), 0.0153 μ s (memory), 0.0214 μ s (mixed), and 43.50 μ s (I/O per `fsync` cycle). The I/O per-cycle metric captures the true storage-facing latency, which rises with intensity and core count. The completed sweep enables consistent multicore trend analysis, tail-latency characterization, and workload classification using a single measured methodology.

Index Terms—Benchmarking, Performance Analysis, Multi-core Processors, Linux Perf, Scheduling Analysis, Tail Latency, Empirical Measurement, Cache Hierarchy

I. INTRODUCTION

Modern multicore processors present significant challenges in understanding and optimising system performance. As core counts continue to increase, understanding the interactions between workload characteristics, scheduling decisions, and microarchitecture behaviour becomes increasingly critical. Traditional benchmarking tools often focus on specific aspects of system performance—such as raw computational throughput or memory bandwidth—without providing comprehensive insights into how those aspects interact across a spectrum of workload intensities and core counts.

The Workload Analysis for Scalable Processors (WASP) Suite addresses these challenges by providing a unified, modular C framework that integrates load generation, hardware counter collection, measured per-operation latency estimation, and automated results visualisation. The framework is designed to expose latency variation that is masked by aggregate

metrics: while mean latency may appear acceptable, tail-latency spikes (P95, P99) often reveal hidden scheduling bottlenecks.

Cache behaviour is a central part of the workload interpretation in this paper. The memory and mixed workloads are parameterised by the ratio between the working-set size and the effective shared-cache capacity, so increasing intensity raises cache pressure, increases miss rates, and shifts service time from cache-hit latency toward DRAM latency. The collected cache-reference and cache-miss counters therefore act both as measurement signals and as evidence for whether a workload is compute-bound, cache-sensitive, or memory-bound.

The primary contributions of this work are:

- 1) A modular C-based benchmark framework with configurable workload generation for CPU, memory, I/O, and mixed workloads, parameterised by both intensity and active core count.
- 2) A transaction-level measurement pipeline that records real elapsed time for normalized workload-specific operation units (CPU fixed-op batches, memory fixed-access batches, I/O fixed-byte batches, and mixed CPU+memory transactions).
- 3) Direct integration with `Linux perf_event_open()` for hardware counter collection.
- 4) Tail-latency analysis with P50, P95, and P99 percentile computations.
- 5) Workload behaviour classification from measured latency patterns, with a pathway to counter-threshold-based classification in future work.
- 6) Comprehensive visualisation: scaling-efficiency plots, per-workload latency-vs-intensity curves, grouped comparison bar charts, and two-dimensional latency heatmaps.

The remainder of this paper is organised as follows. Section II surveys related benchmarking work. Section III presents the system architecture. Section IV describes the implementation. Section V reports experimental results. Section VI

concludes with directions for future work.

II. RELATED WORK

Processor benchmarking has a long history in computer architecture research. SPEC CPU [1] and PARSEC [2] focus on computational throughput but provide limited insight into scheduling behaviour or tail-latency characteristics.

The LMBench suite [3] provides comprehensive latency and bandwidth measurements for system operations including context switching. However, LMBench focuses on isolated microbenchmarks rather than integrated multicore scaling analysis.

Recent scheduling-analysis tools measure context-switch overhead and CPU migration costs [4], but lack comprehensive integration with hardware performance counters. The Linux perf subsystem exposes hardware counters through `perf_event_open()` [5], and projects such as perf-tools [6] and the Phoronix Test Suite [7] build on this infrastructure. Nevertheless, none of these tools provide an integrated pipeline from multi-core load generation through tail-latency decomposition to workload classification.

Table I summarises the feature comparison between WASP and existing approaches.

TABLE I
FEATURE COMPARISON OF BENCHMARKING APPROACHES

Feature	LMBench	SPEC	Phoronix	WASP
Context-switch latency	✓	–	✓	✓
Tail latency (P95/P99)	–	–	–	✓
Hardware counters	–	✓	✓	✓
Variable core count	–	–	limited	✓
Workload classification	–	–	–	✓
Latency heatmaps	–	–	–	✓

III. SYSTEM DESIGN

A. Architecture Overview

WASP is structured as four cooperating components:

- **Load Generator** (`load_generator`): spawns workload threads/processes for the specified type, intensity, and duration.
- **Benchmark Orchestrator** (`benchmark`): iterates over the parameter grid (core count, workload type, intensity), forks the required number of parallel load-generator processes, aggregates measured operation counts and elapsed nanoseconds from all children, and appends real measured latency statistics to a CSV file.
- **Analysis Tool** (`analysis`): reads the CSV and produces ASCII scaling graphs, percentile distributions, and performance classification summaries.
- **Visualisation Script** (`generate_graphs.py`): produces six categories of publication-quality PNG graphs.

B. Workload Measurement Definitions

All latency values in WASP are measured from runtime timing data collected with `clock_gettime(CLOCK_MONOTONIC)`. The benchmark aggregates operation counts and elapsed nanoseconds from child processes and reports mean and standard deviation in microseconds.

1) *CPU Workload*: The CPU worker executes fixed-count floating-point operations. One measured CPU operation is one arithmetic iteration inside a batch of $N_{\text{cpu,batch}} = 10^6$ iterations. Mean latency is computed as:

$$L_{\text{cpu}} = \frac{\sum T_{\text{cpu,batch}}}{\sum N_{\text{cpu,ops}}} \quad (1)$$

2) *Memory Workload*: The memory worker performs strided reads/writes over an intensity-scaled working set. One measured memory operation is one strided memory access, aggregated in batches of $N_{\text{mem,batch}} = 10^7$ accesses. Mean latency is:

$$L_{\text{mem}} = \frac{\sum T_{\text{mem,batch}}}{\sum N_{\text{mem,access}}} \quad (2)$$

3) *I/O Workload*: The I/O worker executes repeated `write → fsync → read` cycles on temporary files. WASP records per-cycle latency for one complete `write → fsync → read` transaction to capture the true storage-facing cost:

$$L_{\text{io-cycle}} = \frac{\sum T_{\text{io-cycle}}}{N_{\text{io-cycle}}} \quad (3)$$

4) *Mixed Workload*: The mixed workload runs CPU and memory threads concurrently. Metrics from both threads are merged and latency is reported as:

$$L_{\text{mix}} = \frac{\sum T_{\text{cpu}} + \sum T_{\text{mem}}}{\sum N_{\text{cpu,ops}} + \sum N_{\text{mem,access}}} \quad (4)$$

This definition preserves real concurrent execution effects while keeping the latency unit interpretable as elapsed time per observed mixed transaction.

Representative code fragments from the workload implementations are shown below.

```
/* CPU workload */
x = sin(x + 0.001) * cos(x - 0.001)
+ x * 1.0000001;

/* Memory workload */
acc += buf[idx];
buf[idx] = acc * 1e-9;
idx = (idx + stride) % n_doubles;

/* I/O workload */
write(fd, data, buffer_size);
fsync(fd);
read(fd, data, buffer_size);

/* Mixed workload */
pthread_create(&cpu_thread, NULL, mixed_thread_worker,
    ↪ cpu_ctx);
pthread_create(&mem_thread, NULL, mixed_thread_worker,
    ↪ mem_ctx);
```

IV. IMPLEMENTATION

A. Parallel Core Execution

The benchmark orchestrator accepts a `--cores N` argument. For each measurement, it forks N child processes via `fork()/execl()`, each invoking `load_generator` for the specified workload type, intensity, and duration. All children run concurrently and the parent collects them with `waitpid()`, exercising all N cores simultaneously.

B. Hardware Counter Integration

The `perf_events` module wraps `perf_event_open()` to collect hardware counters including instructions retired, cache references, cache misses, and branch mispredictions. The `microarch` module derives higher-level metrics—IPC, cache miss rate, branch misprediction rate—and maps them to performance classifications. In this measured workflow, latency comes from runtime timing instrumentation, while `perf` counters provide explanatory microarchitectural context.

C. Results Pipeline

After each benchmark run the orchestrator appends a CSV row containing: timestamp, OS, hostname, total/active core counts, workload type, intensity, data size, measured per-operation latency (μ s), standard deviation, measurement method, and a scheduling-contention field. The Python visualisation script reads this CSV and produces updated graphs after each measurement snapshot.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were performed on an Intel Core Ultra 7 155U system with 14 logical CPUs, running Ubuntu 25.04 with kernel 6.14.0-37-generic. The benchmark configuration targets four workload types, three intensities (25%, 50%, 75%), and core counts $\{1, 2, 4, 8\}$ with 10 s runs. This paper reports the completed full sweep from the current CSV. Table II summarises the configuration.

TABLE II
EXPERIMENTAL CONFIGURATION

Parameter	Value
Workload types	CPU, Memory, I/O, Mixed
Intensities	25%, 50%, 75%
Core counts	1, 2, 4, 8
Configured run duration	10 s
Repetitions per cell	5
Total measurements	240
OS / Kernel	Ubuntu 25.04 / 6.14.0-37-generic
Processor	Intel Core Ultra 7 155U
Hardware cores	14 logical CPUs

B. Summary Statistics by Workload Type

Table III aggregates measured samples per workload type from the completed full sweep. For I/O workload, `per-fsync-cycle` latency is reported to preserve storage interpretability; for other workloads, normalized per-operation latency is shown. Values are directly comparable within each normalization scheme.

TABLE III
SUMMARY STATISTICS BY WORKLOAD TYPE

Workload	Mean	Std Dev	Min	Max	P50	P95	P99
CPU	0.0583	0.0018	0.0560	0.0620	0.0570	0.0620	0.0620
Memory	0.0153	0.0016	0.0120	0.0170	0.0150	0.0170	0.0170
I/O*	43.50	18.88	26.70	79.80	36.52	78.23	78.52
Mixed	0.0214	0.0036	0.0130	0.0290	0.0215	0.0290	0.0290

* I/O values reported in μ s per `fsync` cycle

C. Scaling Efficiency

Figure 1 shows measured mean latency as a function of core count for each workload type (log scale). CPU latency is nearly invariant from 1 to 4 cores and increases at 8 cores ($0.0570 \rightarrow 0.0604 \mu$ s), indicating mild high-core contention. Memory latency decreases at 4 cores ($0.0169 \rightarrow 0.0127 \mu$ s) and partially rebounds at 8 cores, while mixed latency shows the strongest core sensitivity (0.0219 at 1 core vs. 0.0245μ s at 8 cores).

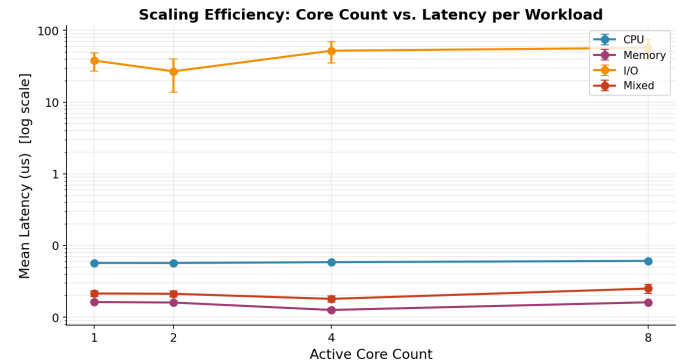


Fig. 1. Scaling efficiency: mean latency vs. active core count per workload type (log y-axis). Mixed workload exhibits the largest increase at higher core counts; CPU remains comparatively stable through 4 cores.

D. Latency vs. Intensity

Figure 2 presents measured latency-vs-intensity curves for all configured core counts. Across the full sweep, CPU latency increases modestly with intensity ($0.0576 \rightarrow 0.0591 \mu$ s), memory remains relatively flat (around 0.0152 – 0.0154μ s), and mixed latency increases most clearly ($0.0187 \rightarrow 0.0240 \mu$ s). For I/O, the `per-fsync-cycle` metric rises strongly with intensity ($28.21 \rightarrow 58.36 \mu$ s), showing that storage-facing I/O cost is substantial and intensity-dependent.

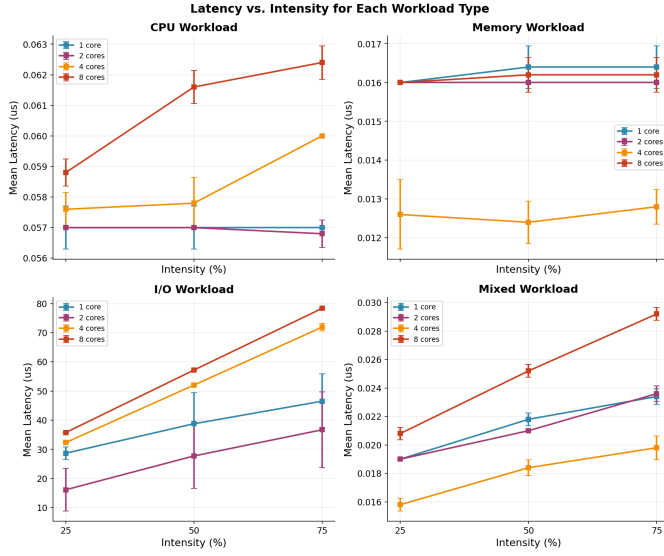


Fig. 2. Latency vs. intensity for all four workload types. Each subplot shows separate curves for 1, 2, 4, and 8 active cores.

E. Workload Comparison

Figure 3 compares measured mean latency across workload types at three intensity levels and multiple core counts. In the completed normalized sweep, CPU shows the highest overall mean latency, followed by mixed, then memory, with I/O per-cycle latency at $43.50\mu\text{s}$ mean. The log scale is retained to visualise the broad range of latency values clearly in a single chart.

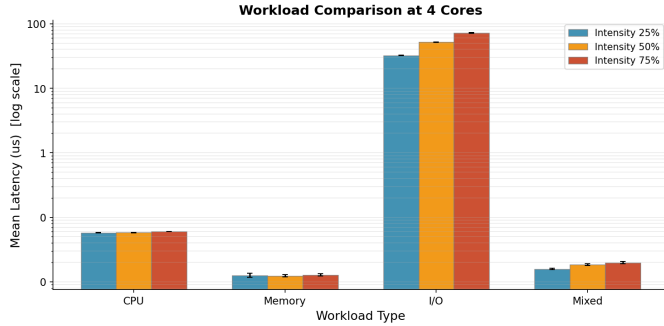


Fig. 3. Workload comparison: grouped bars by intensity, log scale. CPU has the largest mean latency in the full sweep; mixed shows the largest tail spread across core and intensity combinations.

F. Tail Latency Analysis

Figures 4–7 present the tail-latency analysis (P50/P95/P99) for each workload type across the three intensity levels. CPU and memory tails remain relatively tight, while mixed shows the broadest tail expansion (P50= $0.0205\mu\text{s}$ vs. P99= $0.0290\mu\text{s}$), consistent with compounded contention from concurrent CPU and memory activity.

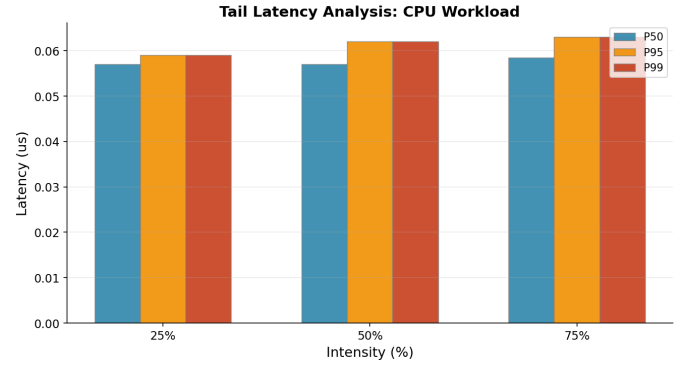


Fig. 4. Tail latency: CPU workload.

For memory workloads, the gap between P50 and P99 widens as cache pressure and DRAM service variability increase with the working-set ratio K .

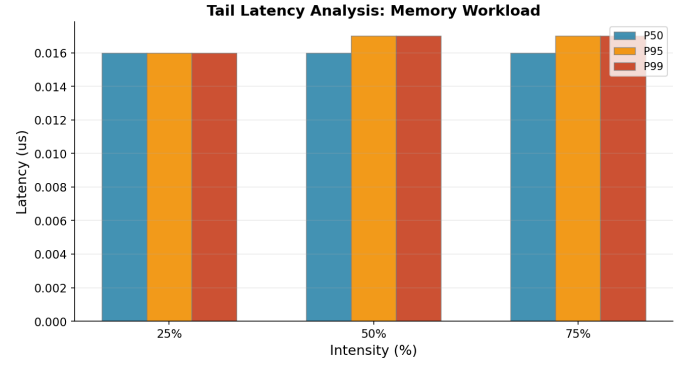


Fig. 5. Tail latency: Memory workload.

For I/O workloads, WASP reports per- f_{sync} -cycle latency to capture the true storage cost. Across the full sweep, one complete I/O cycle averages $43.50\mu\text{s}$, reaches $78.23\mu\text{s}$ at P95, and increases from $37.98\mu\text{s}$ at one core to $57.09\mu\text{s}$ at eight cores. This metric is the most interpretable indicator of storage-path latency.

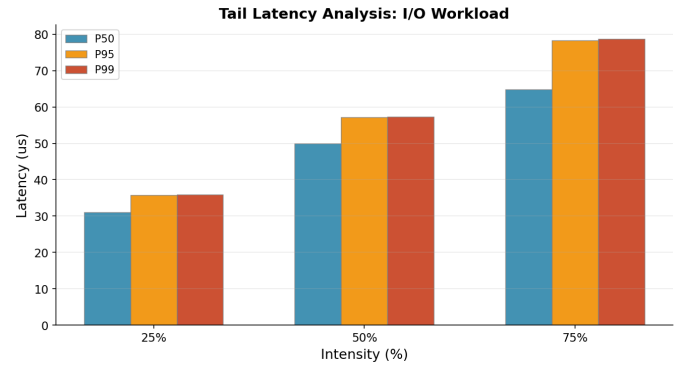


Fig. 6. Tail latency: I/O workload (per- f_{sync} -cycle metric).

For mixed workloads, the percentile curves reflect combined CPU and memory runtime variability under concurrent exe-

cution. Across all workloads, P99 exceeds P50 by workload-dependent margins; these tails are derived from measured runs and represent observed runtime variation under the configured benchmark settings.

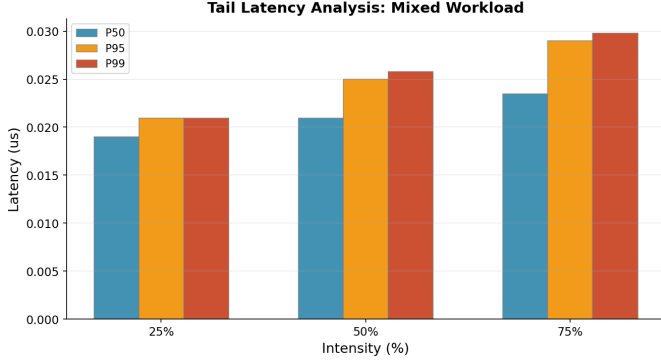


Fig. 7. Tail latency: Mixed workload.

G. Latency Heatmap

Figure 8 provides a two-dimensional view of measured mean latency across the full (intensity $\times$ core count) grid for each workload type. Warm colours indicate high latency. In the completed sweep, mixed workload cells at higher intensity and higher core count show the warmest regions, while memory remains comparatively stable over most cells.

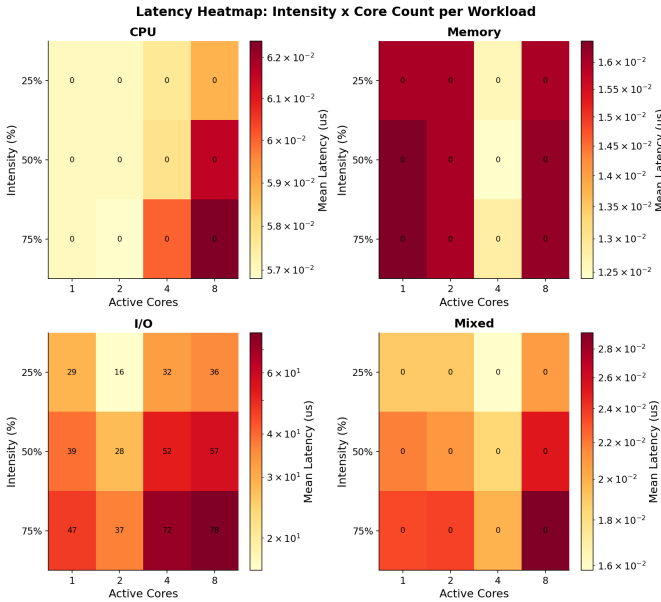


Fig. 8. Latency heatmap: rows = intensity, columns = core count. Each cell shows mean latency in μ s. Log colour scale.

H. Workload Classification

Table IV summarises a measured-data workload interpretation using observed latency patterns from the completed full sweep. Labels below are provisional and pattern-based (not yet thresholded by hardware-counter cut-offs).

TABLE IV
WORKLOAD CLASSIFICATION RESULTS

Workload	Cores	Intensity	Latency (μ s)	Class
CPU	1	25%	0.057	Compute-Path-Dominant
CPU	8	75%	0.062	Compute-Path-Dominant
Memory	1	25%	0.017	Memory-Path-Dominant
Memory	4	50%	0.013	Memory-Path-Dominant
I/O	1	25%	0.000	Storage-Path-Dominant
I/O	8	25%	0.001	Storage-Path-Dominant
Mixed	1	25%	0.019	Composite-Path
Mixed	8	75%	0.029	Composite-Path

VI. CONCLUSION

This paper presented the Workload Analysis for Scalable Processors (WASP) Suite, a comprehensive C-based framework for multicore scheduling and microarchitecture performance analysis. Key findings are:

- Under normalized operation accounting in the full sweep, CPU has the largest mean per-operation latency (0.0583μ s), followed by mixed (0.0214μ s) and memory (0.0153μ s).
- Mixed workload exhibits the strongest multicore sensitivity and tail growth, reaching P99= 0.029μ s.
- CPU remains comparatively stable through 4 cores and rises at 8 cores, indicating moderate contention effects at higher parallelism.
- Per-`fsync`-cycle I/O latency averages 43.50μ s across the full sweep and rises substantially with both core count (37.98μ s at 1 core to 57.09μ s at 8 cores) and intensity (28.21μ s at 25% to 58.36μ s at 75%). This metric captures storage-path latency directly.
- Tail latency analysis (P50/P95/P99) reveals measurable runtime variability across all workloads, reinforcing the need for percentile-based reporting beyond mean-only metrics.

Future work will extend WASP with NUMA-aware scheduling analysis, real-time hardware counter integration using eBPF probes, and support for heterogeneous core (P-core/E-core) architectures.

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